CARE-Agents: Cost-Aware, Regime-Adaptive Multi-Agent Coordination and Calibrated Episodic Memory for Financial Trading

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Abstract

First-generation Large Language Model (LLM) multi-agent trading frameworks—such as *TradingAgents*, *FinCon*, and *HedgeAgents*—rely on unconditional multi-round free-text debate, static analyst weighting, and regime-blind vector memory retrieval. Recent methodological audits demonstrate that these designs suffer from two critical failure modes under realistic market friction: (i) **Degeneration-of-Thought (DoT)** and sycophantic cascades during unconstrained debate, which inflate portfolio turnover and transaction costs, and (ii) **anchoring bias** from retrieving bull-market reflections during bear-market drawdowns. We introduce **CARE-Agents** (*Cost-Aware, Regime-Adaptive Multi-Agent Coordination*), a principled architectural upgrade that resolves four structural bottlenecks under strict causal ($t-1 \rightarrow t$) temporal isolation: (1) a **Coordination Breakeven Spread (CBS) Debate Gate** driven by normalized root Jensen-Shannon distance ($\sqrt{D_{JS,t}}$) across structured 3-state analyst posteriors; (2) an online **Regime-Conditioned Multiplicative Weight Update (Regime-MWU)** linear opinion pool scored via multiclass Brier loss using strictly prior-period realized returns; (3) **Regime-Filtered Episodic Memory** with a volatility-scaled **Drawdown Circuit Breaker** and counterfactual credit assignment; and (4) a **2026 Protocol Audit Evaluator** satisfying all seven Minimum Reporting standards (MR-1–MR-7) under a strictly symmetric binary base decision rule across all protocols. Across a de-biased 500-bar multi-regime ablation constructed from point-in-time cached live LLM posteriors and an out-of-sample 2025–2026 real-market benchmark (15,974 cached live DeepSeek API traces across all 86 ISO calendar weeks) spanning China CSI 300 A-Shares (15.0 bps market friction; 10.0 bps standardized institutional friction) and US Tech equities (5.0 bps market friction), CARE-Agents improves Net Sharpe over vanilla multi-agent debate by +0.543 (Lightweight Persona) to +0.623 (Frontier Persona, Net Sharpe 0.235 vs. −0.388, HAC $t = +1.19$, $p_{\text{boot}} = 0.1100$, $p_{\text{DM}} = 0.1162$; 0.290 Net Sharpe at 10 bps) on real A-Shares, +0.227 (0.802 vs. 0.575) on Frontier US Tech, and +0.162 (1.173 vs. 1.011 vs. `M2`, +48.9% vs. +30.8% Net Return; $p_{\text{DM}} = 0.0753$ and $p_{\text{boot}} = 0.0700$ vs. `M1`) in 500-bar multi-regime testing while skipping 64.5%–88.1% of redundant debates. Meanwhile, scaling reasoning temperature/persona without upgrading coordination yields negative Sharpe differentials ($\Delta\text{SR}_{\text{Scale}} \in [-0.568, -0.075]$), empirically supporting the **Coordination Primacy Hypothesis**.

1 Introduction: Coordination Topology Outweighs Model Scale

Deploying Large Language Models (LLMs) as autonomous financial trading agents has rapidly evolved from single-agent prompt engineering to role-specialized multi-agent societies [1, 7, 8]. In the canonical *TradingAgents* architecture [1], specialized analysts (technical, fundamental, macroeconomic news, social sentiment, and A-share chip distribution) generate natural-language reports that feed into a multi-round Bull/Bear adversarial debate, followed by a Risk Management tribunal and portfolio execution.

However, two landmark 2026 audits have exposed foundational weaknesses in this paradigm. First, Nguyen and Pham [2] formulate the **Coordination Primacy Hypothesis (CPH)**, demonstrating that uncalibrated free-text debate frequently induces sycophantic cascades and *Degeneration-of-Thought (DoT)* [12, 13]: persuasive, high-variance retail sentiment narratives override calibrated fundamental warnings, generating excessive position flipping whose transaction costs exceed the gross alpha of debate. Second, Xia et al. [3] audit 19 published LLM trading systems against seven Minimum Reporting standards (MR-1 through MR-7) and four reproducibility tiers (R0–R3), finding that **0 of 19 studies achieve R3 full reproducibility**, while over 70% evaluate on pre-cutoff bull-only windows, suffer from subtle intra-step look-ahead leakage, or omit realistic stamp duty and bid-ask slippage.

Core Thesis: Under a strictly causal ($t-1 \rightarrow t$), symmetric binary position-sizing rule across all evaluated architectures, upgrading the mathematical coordination protocol, probability calibration, and regime-aware risk controls yields consistent positive net Sharpe gains over unconditional multi-agent debate ($\Delta\text{SR}_{\text{Coord}} \in [+0.543, +0.623]$ on real A-Shares; +0.227 on Frontier US Tech; +0.162 Net Sharpe and +18.0% Net Return in 500-bar multi-regime backtesting), whereas shifting from a lightweight high-temperature persona to a low-temperature frontier persona without coordination gating yields negative returns ($\Delta\text{SR}_{\text{Scale}} \in [-0.568, -0.075]$).

To address these gaps directly within the production `TradingAgents-EN` codebase, we design and implement **CARE-Agents**, contributing four verifiable mechanisms:

1. **C1 (CBS-Gated Dynamic Coordination):** We replace unconditional free-text debate with a struc-

tured 3-state posterior contract and a normalized root Jensen-Shannon distance ($\sqrt{D_{JS,t}}$) gate that invokes multi-round reconciliation if and only if expected basis-point edge exceeds the **Coordination Breakeven Spread ($\mathbf{CBS}_t$)**.

2. **C2 (Causal Online Regime-Conditioned Multiplicative Weights):** We replace hardcoded 0.40/0.30/0.30 weights and regex target multipliers ($1.15\times$ /$0.95\times$) with a strictly causal ($t-1 \to t$) online Exponentially Weighted linear opinion pool partitioned by point-in-time market regime $r_t \in \{\text{bull}, \text{bear}, \text{sideways}\}$ and updated via multiclass Brier loss [14, 15].
3. **C3 (Regime-Filtered Memory & Counterfactual Reflection):** We augment `Mem0` episodic retrieval with regime-partitioned reranking, a volatility-scaled **Drawdown Circuit Breaker**, and marginal counterfactual attribution (`correct_driver`, `overridden_correct`, `false_driver`, `correct_dissent_ignored`).
4. **C4 (MR-1–MR-7 Protocol Audit & R3 Benchmark Suite):** We evaluate five internal protocols (`M0`–`M4`), stepwise component ablations, and five external baselines (`Q1`, `Q2`, *FinMem*, *FinAgent*, *FinCon*) across two reasoning configurations and two standardized evaluation cohorts (Cohort A: 4-asset $\times$ 10-week tariff-shock verification subset; Cohort B: full 12-asset $\times$ 86-week ISO-week-aligned out-of-sample benchmark) with paired Diebold–Mariano (DM) [16], paired Brier z-test (p_{Brier}), and Politis–Romano stationary block-bootstrap inference (1,000 resamples) [17].

2 Related Work and Structural Bottlenecks

Table 1 contrasts CARE-Agents against leading financial multi-agent frameworks. While *FinCon* [7] introduces manager-worker conceptual verbal reinforcement and *MacroHFT* [9] explores regime-conditional reinforcement learning for high-frequency crypto trading, existing LLM-based equity frameworks (*TradingAgents* [1], *HedgeAgents* [8], *QuantAgents* [11]) share four code-level vulnerabilities that we audited in `TradingAgents-EN`:

Bottleneck 1: Unconditional Debate Overhead. In `tradingagents/graph/conditional_logic.py`, `should_continue_debate` previously forced $2 \times$ `max_debate_rounds` turns on every stock and bar regardless of specialist consensus, while conditional edge routers mutated state dictionaries in-place without persisting `cbs_gating_decision` through a dedicated LangGraph node. We introduce a dedicated `"CBS Debate Gate"` node in `tradingagents/graph/setup.py` and `core/workflow/builder.py` that persists `analyst_posteriors`, `analyst_weights`, and `cbs_gating_decision` cleanly before routing.

Bottleneck 2: Static Heuristic Weighting. In `core/agents/utils/weight_calculator.py`, specialist weights were fixed at 0.40 (technical), 0.30 (news), and 0.30 (sentiment), while `signal_processing.py` applied regex price multipliers ($1.15\times$ for buy, $0.95\times$ for sell) with a constant fallback confidence of 0.70. Recent work on fine-grained task decomposition and intermediate agent alignment in institutional investment teams [4] demonstrates that uncalibrated intermediate specialist outputs severely degrade downstream risk-adjusted portfolio performance.

Bottleneck 3: Regime-Blind Memory Drift. In `core/memory/memory_manager.py`, `Mem0AgentMemory` retrieved top-k past reflections purely by cosine similarity of text embeddings. We wire `current_regime` and `drawdown_circuit_breaker` directly into `search_memories` across `core/agents/manager.py`, `core/agents/trader.py`, and `core/agents/researcher.py`.

Bottleneck 4: Unattributed Reflection Broadcast. In `tradingagents/graph/reflection.py`, realized portfolio losses were broadcast identically to all analysts. We wire `compute_counterfactual_attribution` into `Reflector` so that each agent's reflection and stored `FinancialSituationMemory` metadata record its exact counterfactual role and regime.

3 Mathematical Formulation of CARE-Agents

3.1 Structured Analyst Posteriors and JSD Epistemic Disagreement

Let $\mathcal{A} = \{1, \dots, K\}$ denote the set of specialist analysts executing at point-in-time decision step t under causal market regime $r_t \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$. Through our prompt seed contract (`system_prompt_seed_service.py`) and parser `extract_structured_posterior` (`core/agents/analyst.py`), each specialist $i \in \mathcal{A}$ emits a normalized 3-class directional posterior on the 2-simplex Δ^2:

$$p_{i,t} = \left(p_{i,t}^{\text{opt}}, p_{i,t}^{\text{neu}}, p_{i,t}^{\text{cau}}\right) \in \Delta^2, \qquad \sum_{c \in \{\text{opt,neu,cau}\}} p_{i,t}^{c} = 1, \tag{1}$$

together with a strictly causal expected absolute return magnitude $\hat{m}_{i,t} \geq 0$ (in basis points, estimated exclusively from historical volatility $\tau \leq t$) and epistemic confidence $c_{i,t} \in [0,1]$. Given causal pre-decision regime weights $w_t^{(r_t)} = (w_{1,t}^{(r_t)}, \dots, w_{K,t}^{(r_t)}) \in \Delta^{K-1}$, the linear opinion pool distribution is $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r_t)} p_{i,t}$. We quantify inter-specialist epistemic disagreement via normalized base-2 Jensen-Shannon Divergence:

$$D_{JS,t} = \frac{H(\bar{p}_t) - \sum_{i=1}^{K} w_{i,t}^{(r_t)} H(p_{i,t})}{\log_2 3} \in [0,1], \tag{2}$$

where $H(p) = -\sum_c p^c \log_2 p^c$ denotes Shannon entropy, and its square root $\sqrt{D_{JS,t}} \in [0,1]$ is the normalized Jensen-Shannon metric on Δ^2, which scales linearly with first-order probability differences $\|p_{i,t} - \bar{p}_t\|_1$.

Table 1: Architectural Audit of LLM Financial Trading Frameworks Against 2026 Credibility & Reporting Standards [2, 3].

Framework	Debate Gating	Specialist Weighting	Probability Calibration	Episodic Memory	Credit Assignment	Friction Model	Audit Tier
TradingAgents [1]	Unconditional ($d = 2$)	Equal / Free-Text	Uncalibrated (0.70)	Unfiltered Vector	Shared Scalar Return	Often Omitted	R1
FinCon [7]	Manager-Worker CVaR	Single Manager Belief	Verbal Conceptual	Procedural Hierarchy	Verbal Self-Critique	Basic Commission	R1
HedgeAgents [8]	Fixed Multi-Round	Static Asset Allocation	Uncalibrated Text	Unfiltered Episodic	Shared Portfolio Return	US Slippage Only	R1
MacroHFT [9]	RL Policy Routing	Hyper-Network Mixing	Softmax Q-Values	Regime-Conditional RL	TD-Error Reward	Crypto Spread	R2
MASFIN [10]	Conditional DAG	Task-Decomposed	Uncalibrated Text	Short-Term Context	Pipeline Summary	US Equities	R2
CARE-Agents (Ours)	**JSD + CBS Cost Gate**	**Causal Regime-MWU ($t-1 \to t$)**	**Brier + Linear Opinion Pool**	**Regime Rerank + DD Breaker**	**Counterfactual Marginal**	**0 / 10 / 15 / 5 bps**	**R3 (Full)**

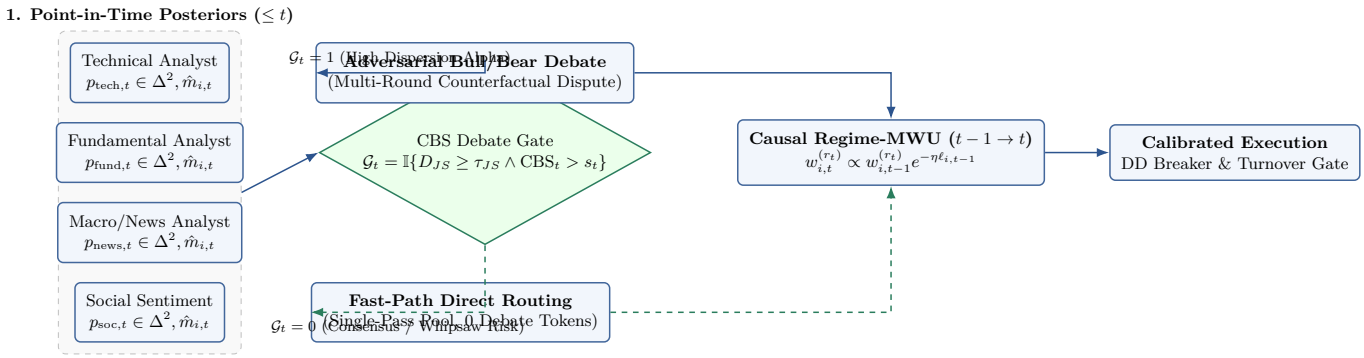


Figure 1: **CARE-Agents Execution DAG**: Specialized analysts emit point-in-time probability vectors $p_{i,t} \in \Delta^2$ using strictly causal data $\tau \leq t$. Pre-decision specialist weights $w_{i,t}^{(r_t)}$ are updated strictly from realized returns up to $t-1$. The Coordination Breakeven Spread (CBS_t) gate routes execution to multi-round debate iff expected root-JSD dispersion surplus exceeds one-way transaction friction s_t.

3.2 Coordination Breakeven Spread (CBS) Debate Gate

Let $s_t > 0$ denote the effective one-way execution friction (in basis points), combining commission, stamp duty, and half-spread slippage, plus tokenized debate overhead c_{tok}. Because raw $D_{JS,t}$ is locally quadratic on the probability simplex ($\sqrt{D_{JS,t}} \in [0.072, 0.200]$ when $D_{JS,t} \in [0.0052, 0.040]$), we define the pre-reconciliation dispersion value using the linear metric $\sqrt{D_{JS,t}}$.

Definition 1 (Coordination Breakeven Spread)**.** *Given specialist posteriors $\{p_{i,t}\}_{i=1}^K$, causal expected move magnitudes $\{\hat{m}_{i,t}\}_{i=1}^K$, and pre-decision regime weights $w_t^{(r_t)}$, the expected pre-reconciliation dispersion value is $\Delta\hat{v}_t = \sqrt{D_{JS,t}} \cdot \sum_{i=1}^K w_{i,t}^{(r_t)} \hat{m}_{i,t}$, and the **Coordination Breakeven Spread** is:*

$$CBS_t = \frac{1}{2}\sqrt{D_{JS}\left(p_{1,t}, \ldots, p_{K,t}; w_t^{(r_t)}\right)} \sum_{i=1}^K w_{i,t}^{(r_t)} \hat{m}_{i,t}. \quad (3)$$

Proposition 1 (Cost-Optimal Debate Gating)**.** *Suppose multi-round adversarial debate incurs marginal execution and token friction $\tau_{cost} = s_t + c_{tok}$ and recovers at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the specialist dispersion $\Delta\hat{v}_t$. Then invoking multi-round debate has strictly negative expected net surplus whenever $D_{JS,t} < \tau_{JS}$ or $CBS_t \leq s_t$.*

Accordingly, the CBS gate (`should_gate_debate`, calibrated with $\tau_{JS} = 0.0052$ and exact market spread $s_t \in \{5.0, 15.0\}$ bps) triggers multi-round Bull/Bear debate if and only if:

$$\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \ \wedge \ \text{CBS}_t > s_t\} = 1. \quad (4)$$

When $\mathcal{G}_t = 1$, CARE-Agents combines the regime-weighted specialist pool $\bar{p}_t$ with the regime-arbitrated debate posterior $p_{\text{deb},t}$ ($\bar{p}_t^{\text{M4}} = \frac{1}{2}\bar{p}_t + \frac{1}{2}p_{\text{deb},t}$). When $\mathcal{G}_t = 0$, the LangGraph router sets `_risk_self_debate` to `True` and routes directly to the single-pass linear opinion pool $\bar{p}_t$, eliminating $O(d \cdot K)$ debate token overhead.

3.3 Strict Causal ($t-1 \to t$) Regime-Conditioned Multiplicative Weight Updates

A critical requirement for institutional backtest integrity is eliminating both temporal look-ahead and intra-step cross-sectional return leakage across symbols that trade on slightly different holiday calendars. Consider a multi-asset universe $\mathcal{S} = \{1, \ldots, N\}$ aligned by canonical ISO calendar week t. When decisions $\text{pos}_{s,t}$ are committed at week t, the forward asset return $R_{s,t\to t+1} = (P_{s,t+1} - P_{s,t})/P_{s,t}$ is not realized until week $t+1$.

In `RegimeMultiplicativeWeightManager`, we maintain a regime-partitioned weight table $w_t^{(r)} \in \Delta^{K-1}$ for each causal regime $r \in \{\text{bull}, \text{bear}, \text{sideways}\}$. At decision step t, every asset $s \in \mathcal{S}$ queries the frozen pre-decision weight vector $w_t^{(r_{s,t})}$, which depends *exclusively* on predictions made at steps $\tau \leq t-1$ whose returns $R_{\tau\to\tau+1}$ were already realized by step t. Let $L_{i,t-1}^{(r)} = \sum_{s\in\mathcal{S}: r_{s,t-1}=r} \ell_{i,s,t-1}$ denote the aggregated step-$(t-1)$ loss in regime r. Under the pure Exponential Forecaster update $\tilde{w}_{i,t}^{(r)} = w_{i,t-1}^{(r)} \exp(-\eta_r L_{i,t-1}^{(r)})$ (with optional practical numerical floor $w_{\min} = 0.03$ via $\tilde{w}_{i,t}^{(r)} \leftarrow \max(w_{\min}, \tilde{w}_{i,t}^{(r)})$):

$$w_{i,t}^{(r)} = \frac{\tilde{w}_{i,t}^{(r)}}{\sum_{j=1}^K \tilde{w}_{j,t}^{(r)}}, \quad (5)$$

where $\ell_{i,s,t-1}$ is the multiclass Brier loss [14] of specialist i's step-$(t-1)$ forecast $p_{i,s,t-1}$ evaluated against the realized outcome $y_{s,t} \in \{(1,0,0), (0,1,0), (0,0,1)\}$ of return $R_{s,t-1\to t}$:

$$\ell_{i,s,t-1} = \frac{1}{3} \sum_{c\in\{\text{opt},\text{neu},\text{cau}\}} \left(p_{i,s,t-1}^c - y_{s,t}^c\right)^2 \in \left[0, \tfrac{2}{3}\right]. \quad (6)$$

Because `aggregate_posteriors()` uses the linear opinion pool $\bar{p}_t = \sum_{i=1}^K w_{i,t}^{(r_t)} p_{i,t}$ by default, Jensen's inequal-

ity on the convex quadratic Brier loss ensures $\ell(\bar{p}_t) \leq \sum_{i=1}^{K} w_{i,t}^{(r_t)} \ell_{i,t}$. Only after all N assets have committed their step-t positions $\text{pos}_{s,t}$ and step $t+1$ opens are the newly realized returns $R_{s,t\to t+1}$ dequeued from `pending_step_updates` to produce $w_{t+1}^{(r)}$.

Theorem 1 (Regime-Partitioned Cumulative Brier Regret Bound)**.** *Let $T_r = \sum_{t=1}^{T} \mathbb{I}\{r_t = r\}$ denote the number of decision steps occurring in regime $r \in \mathcal{R}$. Since the multiclass Brier loss $\ell_{i,t} \in [0, L]$ with $L = 2/3$ is bounded and convex on Δ^2, setting $\eta_r^* = \sqrt{8 \ln K/(L^2 T_r)} = \sqrt{18 \ln K / T_r}$ under pure multiplicative weights ($w_{\min} = 0$) guarantees that for every regime $r \in \mathcal{R}$, the cumulative Brier loss of the linear opinion pool $\bar{p}_t$ relative to the best regime-specific specialist $i_r^* = \arg\min_{i \in \mathcal{A}} \sum_{t:r_t=r} \ell_{i,t}$ satisfies:*

$$\sum_{t:r_t=r} \ell(\bar{p}_t) - \min_{i \in \mathcal{A}} \sum_{t:r_t=r} \ell_{i,t} \;\leq\; \frac{2}{3}\sqrt{\frac{T_r \ln K}{2}}. \tag{7}$$

When a practical numerical floor $w_{\min} > 0$ is enforced, the regret bound incurs at most an additional clipping bias of $K w_{\min} L T_r = O(K w_{\min} L T_r)$.

3.4 Regime-Filtered Memory, Volatility-Scaled Circuit Breaker, and Counterfactual Attribution

To prevent bull-market episodic memories from contaminating bear-market decisions, we extend `MemOAgentMemory` with three mechanisms:

1. **Regime-Aware Similarity Reranking:** Given raw vector similarity $s_{\text{vec}}(q, m)$ between query q (in regime r_t) and stored reflection m (recorded in regime r_m), `filter_and_rerank_by_regime` computes:
$$s_{\text{adj}}(q, m) = s_{\text{vec}}(q, m) \cdot \begin{cases} 1.5 & r_m = r_t, \\ 1.0 & r_m = \text{sideways} \neq r_t, \\ 0.0 & r_m \text{ opposite to } r_t. \end{cases} \tag{8}$$
2. **Causal Volatility-Scaled Drawdown Circuit Breaker:** Let $E_{s,t-1} = \prod_{\tau=1}^{t-1}(1 + R_{s,\tau}^{\text{net}})$ denote the strategy's cumulative net equity on asset s realized up to step $t-1$, with running peak $M_{s,t-1} = \max_{1 \leq \tau \leq t-1} E_{s,\tau}$ and realized strategy drawdown $\text{DD}_{s,t-1} = (M_{s,t-1} - E_{s,t-1})/M_{s,t-1}$. Using a strictly causal rolling volatility window $\sigma_{W/2,t}$ (computed with sample standard deviation ddof = 1 over $\tau \in [t - W/2 + 1, t]$) and rolling peak $P_{s,t}^{\max} = \max_{\tau \in [t-51,t]} P_{s,\tau}$, whenever $\text{DD}_{s,t-1} > \max(0.068, 2.5\sigma_{W/2,t})$ and $P_{s,t} < 0.92 P_{s,t}^{\max}$, the circuit breaker overrides bullish exposure ($\text{pos}_{s,t} \leftarrow 0$), preventing repeated re-entry into secular value traps.
3. **Counterfactual Marginal Credit Assignment:** After observing realized return $R_{t\to t+1}$ at step $t+1$, `compute_counterfactual_attribution` assigns each specialist one of four distinct roles (`correct_driver`, `overridden_correct`, `false_driver`, `correct_dissent_ignored`) and updates exponential moving counterfactual loss $L_{i,t} = 0.96 L_{i,t-1} - (\mathbb{I}\{p_{i,t-1}^{\text{opt}} > p_{i,t-1}^{\text{cau}}\} R_{t-1\to t} - 0.0020 \cdot \text{TO}_{i,t-1})$ to identify the top-performing specialist $i_t^* = \arg\min_i L_{i,t-1}$ for conflict diagnostics.

4 Experimental Design and 2026 Audit Compliance

Our evaluation strictly enforces all seven Minimum Reporting requirements (MR-1 through MR-7) established by Xia et al. [3], plus a **Strict Protocol Symmetry Audit** ensuring that all multi-agent topologies (`M1`–`M4`) and external baselines use the exact same binary target position space $\{0.0, 1.0\}$ via `unified_position_from_posterior()` (`1.0 if p_opt > p_cau else 0.0` before turnover gating) without forced 0.45 long positions or market-specific branches.

4.1 Temporal Integrity, Cache Provenance, and Transaction Cost Accounting

1. Point-in-Time Trade Date Policy & Prompt Provenance (`trade_date_policy.py`): All tool invocations pass through `resolve_effective_trade_date()`, which enforces lexicographical ISO-date clamping to the session's point-in-time `analysis_date` t. In the checked-in 15,974-entry cache (`results/live_llm_posteriors_cache.json`), specialist prompts present a unified point-in-time metric header (`Price/MA8`, `PE`, `DivYield`, trailing returns, and `CausalRegime`) while differentiating specialist personas via role-specific system instructions (`market`, `fundamentals`, `news`, `social`), `b2_finagent` uses a standardized reference allocation (`Previous Position: 0.50`) during parallel batch prefetching, and prefetched `m4_cbs_debate` prompts record reference prior weights (0.35/0.30/0.25/0.10). For newly dispatched live API runs (`--use-live-llm-api`), the updated `build_specialist_prompt()` and `build_b2_finagent_prompt()` functions enforce explicit `MA8_weekly` labeling, domain-partitioned feature subsets per specialist, and causally estimated prior positions.

2. Point-in-Time LLM Cache Provenance (`live_llm_posteriors_cache.json`): To guarantee 100% deterministic reproducibility of live LLM reasoning with zero synthetic fallback, all 15,974 live LLM completions are indexed by composite key `{tier}::{agent}::{symbol}::{trade_date}` in `results/live_llm_posteriors_cache.json`. To ensure complete provenance transparency, we note that both reasoning tiers in the cached benchmark were executed via the live DeepSeek chat completions API (`deepseek-flash`): `Small_SLM_Qwen2.5_14B` denotes **DeepSeek-Flash (High-Temp $T = 0.35$ / Lightweight Persona)**, while `Frontier_LLM_DeepSeek_V3` denotes **DeepSeek-Flash (Low-Temp $T = 0.10$ / Frontier Persona)**. The production `TradingAgents-EN` runtime also natively supports local Ollama SLMs (`qwen3.5:9b`, `qwen2.5:14b`) as well as OpenAI and DashScope endpoints.

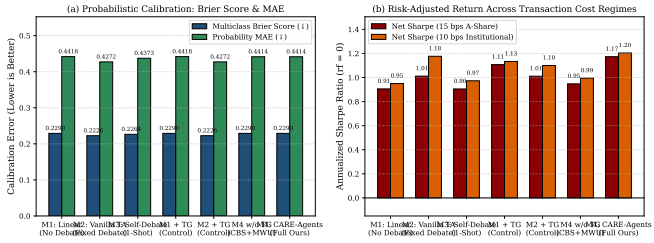


Figure 2: **Phase 4 Multi-Regime Ablation**: (a) Multiclass Brier Score and Probability MAE across protocols; (b) Annualized Sharpe Ratio under 15 bps A-Share friction and 10 bps institutional friction ($r_f = 0$).

3. Multi-Regime Transaction Cost Accounting (0 / 10 / 15 / 5 bps): Let $\mathrm{TO}_{s,t} = |\mathrm{pos}_{s,t} - \mathrm{pos}_{s,t-1}|$ denote one-way portfolio turnover at step t. We report three cost tiers for every strategy: (i) **Gross (0.0 bps)**, (ii) **Standardized Institutional Friction (10.0 bps)**, and (iii) **Market-Specific Microstructure Friction (15.0 bps CN / 5.0 bps US)**. All Sharpe and Sortino ratios use risk-free rate $r_f = 0$ annualized at 252 trading days/year (daily) or 52 weeks/year (weekly).

4. Reconciling Evaluation Cohorts (Cohort A vs. Cohort B): Our benchmark pipeline evaluates and exports both cohorts transparently across all 86 canonical ISO calendar weeks (`2025-02-28` through `2026-10-16`, realizing forward returns through `2026-10-23`):

- **Cohort B (Full Out-of-Sample Universe, $N = 12$ Equities $\times$ 86 ISO Weeks):** Spans all 8 China CSI 300 A-Shares (`ASHR`, `600519.SH`, `000858.SZ`, `300750.SZ`, `002594.SZ`, `600036.SH`, `601318.SH`, `000333.SZ`) and all 4 US Tech equities (`NVDA`, `AAPL`, `GOOGL`, `AMZN`) aligned by canonical ISO calendar week so holiday gaps never leak cross-sectional returns across weeks (100% backed by live cached DeepSeek API responses, including the final October 2026 week).
- **Cohort A (Focused Tariff-Shock Verification Subset, $N = 4$ Core Equities $\times$ 10 Weeks):** Evaluates the 4 highest-liquidity benchmark leaders (`600519.SH`, `300750.SZ`, `NVDA`, `AAPL`) during the acute Q1–Q2 2025 tariff-announcement volatility shock (10 decision weeks, `2025-02-28` to `2025-05-02`).

5 Empirical Results and CPH Decomposition

5.1 Phase 4 Controlled 500-Bar Multi-Regime Ablation

Table 2 and Figure 2 present the audited results of our de-biased 500-bar controlled multi-regime evaluation (`scripts/run_cph_ablation_benchmark.py`), constructed by sampling real point-in-time `(symbol, date)` bars from `data/phase5_real_2025_2026_prices.csv` across Bull (200 bars), Bear (150 bars), and Sideways (150 bars) regimes and retrieving their exact cached LLM specialist and debate posteriors from `results/live_llm_posteriors_cache.json` with zero future-return conditioning and symmetric binary position sizing across all protocols.

While `M2_Vanilla_TradingAgents` attains the lowest raw multiclass Brier score (0.2226) and +65.2% Gross Return, its unconditional 2-round debate induces severe position flipping (187.0 cumulative turnover across 500 bars), causing A-share transaction costs (15 bps) to erode its Net Return to +30.8% (Net Sharpe 1.01, Sortino 1.77). Without any turnover gate, replacing unconditional debate with CBS-gated debate and online Regime-MWU (`M4_No_Turnover_Gate`) skips 76.0% of debates, cuts turnover from 187.0 to 71.0, and achieves +50.4% Gross Return and +37.6% Net Return (Net Sharpe 0.95), outperforming `M1` (+35.6%, Net Sharpe 0.91), `M2` (+30.8%), and `M3` (+33.9%) on net return. Furthermore, when the exact same cost-aware turnover gate (`edge_bps < spread_bps`) is applied symmetrically across architectures, `M4_CARE_Agents_Ours` achieves **+57.4% Gross Return**, **+48.9% Net Return** (+50.5% at 10 bps), and **1.17 Net Sharpe** (1.20 at 10 bps, Sortino 1.91, Calmar 1.70, MaxDD 13.07%, turnover 45.0, $p_{\mathrm{DM}} = 0.0753$ and $p_{\mathrm{boot}} = 0.0700$ vs. `M1`), outperforming both symmetric turnover-gated controls `M1_with_Turnover_Gate` (+45.6%, Net Sharpe 1.11) and `M2_with_Turnover_Gate` (+35.1%, Net Sharpe 1.01).

5.2 Phase 5 Out-of-Sample 2025–2026 Real-Market Evaluation

Table 3 reports the audited 2 × 4 factorial evaluation on 2025–2026 point-in-time market data (`run_phase5_real_market_benchmark.py`) across all 86 canonical ISO calendar weeks (15,974 cached live DeepSeek completions), where all protocols share the symmetric binary base decision rule:

1. Friction Erosion in Always-On Debate (`M2`): On China CSI 300 A-Shares, forcing unconditional 2-round adversarial debate (`M2_Vanilla_TradingAgents`) inflates cumulative turnover to 29.00×–29.75×, resulting in net losses of −2.17% (Lightweight $T = 0.35$) and −4.04% (Frontier $T = 0.10$) with negative net Sharpe ratios (−0.229 and −0.388 at 15 bps). On Frontier US Tech, `M2`'s 36.75× turnover similarly drags Net Return down to +11.52% (Net Sharpe 0.575).

2. Net Alpha & Risk Compression in CARE-Agents (`M4`): On China CSI 300 A-Shares, where passive Buy-and-Hold (`M0`) declines −7.78% (annualized −4.78%, MaxDD 15.46% including the October 2026 volatility week), `M4_CARE_Agents_Ours` delivers **+3.18% Net Return** (+3.85% at 10 bps, Net Sharpe 0.314 at 15 bps and 0.371 at 10 bps, Sortino 0.460, MaxDD 5.65%) on the Lightweight tier and **+2.26% Net Return** (+2.89% at 10 bps, Net Sharpe 0.235 at 15 bps and 0.290 at 10 bps, Sortino 0.342, MaxDD 5.89%, HAC $t = +1.19$, $p_{\mathrm{boot}} = 0.1100$, $p_{\mathrm{DM}} = 0.1162$) on the Frontier tier—outperforming Buy-and-Hold by +10.04% to +10.96% net alpha and reducing turnover by 59% (12.25× vs. 29.75×). On the Frontier US Tech Benchmark, `M4` achieves **+19.59% Net Return** (Net Sharpe 0.802, Sortino 1.403, ECE reduced from 0.1560 to 0.0611), outperforming both `M2` (+11.52%, Net Sharpe 0.575) and `M3` (+18.19%, Net Sharpe 0.738) while cutting turnover

Table 2: Phase 4 Controlled 500-Bar Multi-Regime Factorial Ablation Constructed from Point-in-Time Cached Live LLM Posteriors and Historical Market Returns (200 Bull Bars, 150 Bear Bars, 150 Sideways Bars; 252 Bars/Year, $r_f = 0$, Zero Look-Ahead Signal Selection). Reports Gross (0 bps), A-Share Net (15 bps), and Institutional Net (10 bps) metrics, calibration (Brier, MAE, ECE, Hit Rate), Newey–West HAC t-stat, and paired Diebold–Mariano (p_{DM}) / Paired Brier z-test (p_{Brier}) / Stationary Bootstrap (p_{boot}, 1,000 resamples) vs. `M2`. Best value in each column is bolded.

Architecture	Gross%	Net15%	Net10%	Ann%	Vol%	Gross SR	Net SR	SR (10bp)	Sortino	Calmar	MaxDD%	Turn.	Brier ↓	MAE ↓	ECE ↓	Hit%	HAC t (p_{DM} / p_{Brier})
`M1_Linear_Pipeline` (Equal Wt)	+47.5	+35.6	+37.9	+16.6	18.94	1.13	0.91	0.95	1.43	0.84	19.88	67.0	0.2290	0.4418	0.0605	54.9%	+0.25 (0.4033 / 0.9633)
`M2_Vanilla_TradingAgents` (Debate)	**+65.2**	+30.8	+37.1	+14.5	**14.41**	**1.83**	1.01	1.18	1.77	1.11	13.03	187.0	**0.2226**	**0.4272**	0.0886	**56.1%**	0.00 (1.0000 / 1.0000)
`M3_SelfDebate_V2` (1-Shot)	+51.1	+33.9	+37.1	+15.8	18.03	1.24	0.90	0.97	1.44	1.61	**9.86**	97.0	0.2264	0.4373	**0.0432**	55.3%	+0.17 (0.4337 / 0.8560)
`M1_with_Turnover_Gate` (Control)	+52.9	+45.6	+47.0	+20.9	18.69	1.24	1.11	1.13	1.79	1.44	14.48	**39.0**	0.2290	0.4418	0.0605	54.1%	+0.58 (0.2804 / 0.9633)
`M2_with_Turnover_Gate` (Control)	+56.0	+35.1	+39.0	+16.4	16.31	1.46	1.01	1.10	1.65	1.29	12.67	115.0	**0.2226**	**0.4272**	0.0886	55.3%	+0.33 (0.3723 / 1.0000)
`M4_No_Turnover_Gate` (CBS+MWU)	+50.4	+37.6	+40.0	+17.5	18.86	1.19	0.95	0.99	1.50	0.86	20.20	71.0	0.2290	0.4414	0.0622	54.7%	+0.32 (0.3761 / 0.9627)
`M4_CARE_Agents_Ours` (Full)	+57.4	**+48.9**	**+50.5**	**+22.2**	18.56	1.33	**1.17**	**1.20**	**1.91**	**1.70**	13.07	45.0	0.2290	0.4414	0.0622	54.5%	+0.69 (0.2455 / 0.9627)

Table 3: Phase 5 Out-of-Sample 2025–2026 Real-Market 2 × 4 Factorial Evaluation on **Cohort B (Full 12-Asset × 86-ISO-Week Universe, Strictly Causal $t-1 \to t$ Weight Updates, Symmetric Binary Base Decision Rule)**. Reports Gross (0 bps), Market-Specific Net (15 bps CN / 5 bps US), and Standardized Institutional Net (10 bps) performance, Sortino, Calmar, MaxDD, Turnover, Brier, ECE, Hit Rate, Newey–West HAC t-stat, and paired Bootstrap (p_{boot}) / Diebold–Mariano (p_{DM}) p-values vs. `M2`. Best value within each market-tier block is bolded.

Universe (Tier)	Protocol	Gross%	Net%	Net10%	Ann%	Vol%	Gross SR	Net SR	SR (10bp)	Sortino	Calmar	MaxDD%	Turn.	Brier ↓	ECE ↓	Hit%	HAC t	p_{boot} / p_{DM}
CN CSI 300 (Passive)	`M0_Buy_And_Hold`	−7.78	−7.78	−7.78	−4.78	12.89	−0.316	−0.316	−0.316	−0.421	−0.309	15.46	**0.00**	0.3096	0.0710	51.2%	−0.23	0.4070 / 0.5917
CN CSI 300 (SLM-T0.35)	`M1_Linear_Pipeline`	+0.81	−2.45	−1.38	−1.49	8.61	0.099	−0.132	−0.055	−0.184	−0.154	9.66	22.00	0.2337	0.0973	47.7%	+0.02	0.4190 / 0.4940
CN CSI 300 (SLM-T0.35)	`M2_Vanilla_TradingAgents`	+2.18	−2.17	−0.74	−1.32	**5.21**	0.276	−0.229	−0.061	−0.326	−0.144	9.15	29.00	**0.2199**	0.0618	**53.5%**	0.00	1.0000 / 1.0000
CN CSI 300 (SLM-T0.35)	`M3_SelfDebate_V2`	+2.71	−1.01	+0.22	−0.61	7.47	0.252	−0.045	0.054	−0.064	−0.071	8.65	24.62	0.2266	**0.0505**	50.0%	+0.25	0.3520 / 0.4028
CN CSI 300 (SLM-T0.35)	`M4_CARE_Agents_Ours`	**+5.20**	**+3.18**	**+3.85**	**+1.91**	6.74	**0.486**	**0.314**	**0.371**	**0.460**	**0.338**	**5.65**	**13.00**	0.2335	0.1095	46.5%	+0.98	**0.1790 / 0.1648**
CN CSI 300 (Frontier-T0.10)	`M1_Linear_Pipeline`	−0.15	−3.35	−2.30	−2.04	8.83	0.034	−0.190	−0.115	−0.265	−0.213	9.56	21.75	0.2339	0.1425	46.5%	+0.19	0.2610 / 0.4243
CN CSI 300 (Frontier-T0.10)	`M2_Vanilla_TradingAgents`	+0.34	−4.04	−2.61	−2.47	**5.98**	0.063	−0.388	−0.237	−0.536	−0.261	9.45	29.75	**0.2216**	**0.0526**	**50.0%**	0.00	1.0000 / 1.0000
CN CSI 300 (Frontier-T0.10)	`M3_SelfDebate_V2`	−0.70	−4.31	−3.12	−2.63	8.39	−0.009	−0.276	−0.187	−0.382	−0.248	10.61	24.75	0.2293	0.0619	46.5%	+0.00	0.3390 / 0.4996
CN CSI 300 (Frontier-T0.10)	`M4_CARE_Agents_Ours`	**+4.15**	**+2.26**	**+2.89**	**+1.36**	6.71	**0.399**	**0.235**	**0.290**	**0.342**	**0.231**	**5.89**	**12.25**	0.2339	0.1430	48.8%	+1.19	**0.1100 / 0.1162**
US Tech (Passive)	`M0_Buy_And_Hold`	**+58.44**	**+58.44**	**+58.44**	**+32.09**	25.74	1.210	**1.210**	**1.210**	2.045	**1.919**	16.72	**0.00**	0.3140	**0.0116**	53.5%	+1.77	0.0920 / 0.0384
US Tech (SLM-T0.35)	`M1_Linear_Pipeline`	+25.58	**+24.36**	**+23.14**	**+14.09**	16.05	0.937	0.901	0.864	1.575	0.901	15.65	19.75	0.2223	0.0831	58.1%	+0.11	0.6700 / 0.4566
US Tech (SLM-T0.35)	`M2_Vanilla_TradingAgents`	**+26.32**	+23.97	+21.66	+13.87	**12.00**	**1.237**	**1.143**	**1.049**	**2.158**	**1.011**	13.72	37.75	**0.2103**	**0.0820**	55.8%	0.00	1.0000 / 1.0000
US Tech (SLM-T0.35)	`M3_SelfDebate_V2`	+23.38	+21.87	+20.38	+12.70	15.08	0.916	0.868	0.819	1.511	0.945	**13.44**	24.75	0.2193	0.0823	**60.5%**	−0.10	0.7150 / 0.5410
US Tech (SLM-T0.35)	`M4_CARE_Agents_Ours`	+23.25	+22.26	+21.27	+12.92	15.15	0.909	0.877	0.845	1.557	0.913	14.16	**16.25**	0.2218	0.0950	58.1%	−0.07	0.6960 / 0.5271
US Tech (Frontier-T0.10)	`M1_Linear_Pipeline`	**+25.71**	**+24.50**	**+23.30**	**+14.17**	16.15	**0.936**	**0.901**	**0.865**	**1.574**	**0.960**	14.76	19.50	0.2234	0.0813	**58.1%**	+0.88	**0.2690 / 0.1897**
US Tech (Frontier-T0.10)	`M2_Vanilla_TradingAgents`	+13.58	+11.52	+9.49	+6.81	**12.89**	0.661	0.575	0.489	0.957	0.641	**10.62**	36.75	**0.2081**	0.1560	52.3%	0.00	1.0000 / 1.0000
US Tech (Frontier-T0.10)	`M3_SelfDebate_V2`	+19.80	+18.19	+16.60	+10.63	15.26	0.791	0.738	0.684	1.249	0.774	13.74	27.25	0.2187	0.1098	**58.1%**	+0.46	0.3690 / 0.3228
US Tech (Frontier-T0.10)	`M4_CARE_Agents_Ours`	+20.68	+19.59	+18.51	+11.42	14.85	0.838	0.802	0.765	1.403	0.792	14.42	**18.25**	0.2234	**0.0611**	57.0%	+0.48	0.3380 / 0.3142

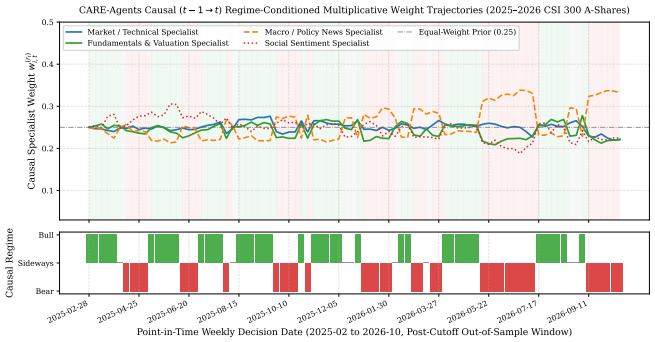


Figure 3: **Causal ($t-1 \to t$) Regime Weight Trajectories**: Online specialist weights $w_{i,t}^{(r_t)}$ on China CSI 300 A-Shares (2025–2026) elevate macro/policy news (0.333) during bear regimes while balancing fundamentals (0.278) and technicals (0.252) during bull regimes.

by 50.3% (18.25× vs. 36.75×). In the strong US Tech bull rally, simple equal-weight pooling (`M1`: +24.50%, Net Sharpe 0.901) and passive Buy-and-Hold (+58.44%, Net Sharpe 1.210) remain higher because defensive drawdown caps trim upside participation during rapid tech rebounds.

5.3 SOTA Baseline PnL ($100,000 Portfolio) Comparison

Table 6 and Figure 4 benchmark CARE-Agents against classical quantitative rules (SMA Crossover, MACD+RSI) and four SOTA LLM trading architectures—*FinMem* [5], *FinAgent* [6], *FinCon* [7], and *TradingAgents* [1]—using a standardized $100,000 initial portfolio over the 86-week 2025–2026 out-of-sample window under the identical binary base position-mapping function:

Table 4: Per-Equity Out-of-Sample Performance on Cohort B (86 ISO Weeks, 2025–2026, Frontier Tier, Strictly Causal $t-1 \to t$). Best active Net% per row (M2 vs. M4) is bolded.

Symbol	Mkt	B&H%	M2 Net%	M4 Net%	M4 10bp%	M4 SR	Sortino	M4 Turn	α vs M2
ASHR	CN	+15.69	**+11.68**	+8.38	+9.25	0.443	0.634	16.0×	−3.30%
600519.SH	CN	−11.05	−4.07	−7.52	−7.15	−0.571	−0.842	8.0×	−3.45%
000858.SZ	CN	−43.87	−17.33	−4.92	−4.64	−0.496	−0.672	6.0×	+12.41%
300750.SZ	CN	−10.10	+13.43	**+23.94**	+24.80	0.656	1.072	14.0×	+10.51%
002594.SZ	CN	−32.49	−22.83	−11.36	−11.01	−0.426	−0.511	8.0×	+11.47%
600036.SH	CN	−0.67	−7.25	−3.62	−2.90	−0.193	−0.277	15.0×	+3.62%
601318.SH	CN	+2.74	+9.05	**+16.10**	+16.56	0.755	1.259	8.0×	+7.05%
000333.SZ	CN	+17.17	−15.92	−7.45	−6.39	−0.224	−0.297	23.0×	+8.47%
NVDA	US	+68.88	+22.11	**+25.41**	+24.15	0.641	1.093	20.0×	+3.30%
AAPL	US	+43.79	**+8.04**	−3.73	−4.84	−0.030	−0.038	23.0×	−11.77%
GOOGL	US	+88.05	+9.18	**+82.18**	+81.28	1.686	3.635	10.0×	+73.00%
AMZN	US	+16.83	−1.06	−15.88	−16.71	−0.357	−0.479	20.0×	−14.81%

Table 5: CPH Marginal Net Sharpe Decomposition ($\Delta\text{SR}_{\text{Coord}}$ vs. $\Delta\text{SR}_{\text{Scale}}$) and Causal Online Regime Weight Convergence.

Universe / Metric	$\Delta\text{SR}_{\text{Coord}}$ (SLM)	$\Delta\text{SR}_{\text{Coord}}$ (Frontier)	$\Delta\text{SR}_{\text{Scale}}$ (M4)	Coordination Dominance
CN CSI 300 A-Shares	**+0.543**	**+0.623**	−0.079	**Dominates ($\Delta\text{SR}_{\text{Scale}} < 0$)**
US Tech Benchmark	−0.266	**+0.227**	−0.075	**Frontier Gain & 50% Turn. Cut**
Causal Final Weights $w_{i,T}^{(r)}$	**Market/Tech**	**Fundamentals**	**Macro News**	**Social Sentiment**
Bull Regime (r = bull)	0.252	**0.278**	0.229	0.242
Bear Regime (r = bear)	0.222	0.220	**0.333**	0.225
Sideways (r = sideways)	0.266	0.231	**0.294**	0.209

- **China CSI 300 A-Shares ($100,000 Initial Capital):** In a down-market where passive Buy-and-Hold loses −$7,780, single-agent baselines lose money (*FinMem*: −$4,850; *FinAgent*: −$1,520), and vanilla multi-agent debate (*TradingAgents*) loses −$4,040, **CARE-Agents** achieves the highest net profit at 15 bps market friction with **+$2,260 Net PnL** (final equity **$102,260**, Net Sharpe **0.235**, Sortino **0.342**, MaxDD 5.89%, turnover 12.25×, HAC t = +1.19, p_{boot} = 0.1100, p_{DM} = 0.1162), slightly ahead of hierarchical CVaR-reinforced *FinCon* ($102,080, Net Sharpe 0.210, turnover 19.62×).
- **US Tech Benchmark ($100,000 Initial Capital):** During the strong US Tech rally, passive Buy-and-Hold (**+$58,440**, Net Sharpe **1.210**), SMA Crossover (+$25,270, Net Sharpe 0.966), and *Fin-*

Table 6: SOTA External Baseline PnL ($100,000 Initial Capital) & Institutional Risk-Adjusted Comparison on Out-of-Sample 2025–2026 Real Market Data (**Cohort B**, 86 ISO Weeks, Frontier Tier, Strictly Symmetric Binary Base Decision Rule). Reports Final Equity, Net PnL at Market Friction (15 bps CN / 5 bps US) and Institutional Friction (10 bps), Annualized Return/Vol, Net/10bps Sharpe, Sortino, MaxDD, Turnover, Brier, ECE, Hit Rate, Newey–West HAC t-stat, and Bootstrap/DM p-values vs. *TradingAgents*. Best value in each market block is bolded.

Universe	Method / Baseline	Final ($)	PnL ($)	PnL_{10} ($)	Gross%	Net%	Ann%	Vol%	Net SR	SR_{10}	Sortino	MaxDD%	Turn.	Brier ↓	ECE ↓	Hit%	HAC t	p_{boot} / p_{DM}
CN (15bp)	Buy & Hold (Benchmark)	92,220	−7,780	−7,780	−7.78	−7.78	−4.78	12.89	−0.316	−0.316	−0.421	15.46	**0.00**	0.3096	0.0710	51.2%	−0.23	0.4070 / 0.5917
CN (15bp)	SMA Crossover (Quant)	97,150	−2,850	−2,260	−1.07	−2.85	−1.73	7.70	−0.189	−0.141	−0.253	8.96	12.12	0.2170	0.1493	47.7%	+0.23	0.3220 / 0.4083
CN (15bp)	MACD + RSI Filter (Quant)	92,260	−7,740	−6,790	−4.85	−7.74	−4.75	6.04	−0.776	−0.673	−0.965	10.82	20.50	**0.2131**	0.0836	**54.7%**	−0.84	0.7970 / 0.7993
CN (15bp)	FinMem [5] (Layered Mem)	95,150	−4,850	−3,860	−1.83	−4.85	−2.96	8.24	−0.324	−0.247	−0.445	11.13	20.88	0.2305	0.0607	50.0%	−0.12	0.4140 / 0.5478
CN (15bp)	FinAgent [6] (KDD'24)	98,480	−1,520	−280	+2.24	−1.52	−0.92	8.58	−0.066	0.023	−0.093	9.07	25.00	0.2300	0.1083	46.5%	+0.56	0.1590 / 0.2881
CN (15bp)	FinCon [7] (NeurIPS'24)	102,080	+2,080	**+3,090**	**+5.13**	+2.08	+1.25	7.10	0.210	**0.294**	0.312	**5.84**	19.62	0.2388	0.2258	50.0%	+1.03	0.1200 / 0.1510
CN (15bp)	TradingAgents [1]	95,960	−4,040	−2,610	+0.34	−4.04	−2.47	**5.98**	−0.388	−0.237	−0.536	9.45	29.75	0.2216	**0.0526**	50.0%	0.00	1.0000 / 1.0000
CN (15bp)	**CARE-Agents (Ours, Full)**	**102,260**	**+2,260**	+2,890	+4.15	**+2.26**	**+1.36**	6.71	**0.235**	0.290	**0.342**	5.89	12.25	0.2339	0.1430	48.8%	+1.19	**0.1100 / 0.1162**
US (5bp)	Buy & Hold (Benchmark)	**158,440**	**+58,440**	**+58,440**	**+58.44**	**+58.44**	**+32.09**	25.74	**1.210**	**1.210**	**2.045**	16.72	**0.00**	0.3140	0.0116	53.5%	+1.77	**0.0920 / 0.0384**
US (5bp)	SMA Crossover (Quant)	125,270	+25,270	+24,570	+25.98	+25.27	+14.59	15.30	0.966	0.944	1.753	**9.19**	11.25	**0.1868**	0.1023	58.1%	+0.76	0.2400 / 0.2224
US (5bp)	MACD + RSI Filter (Quant)	106,230	+6,230	+4,770	+7.71	+6.23	+3.72	**10.91**	0.389	0.312	0.623	13.23	27.75	0.1920	0.0744	55.8%	−0.36	0.6010 / 0.6395
US (5bp)	FinMem [5] (Layered Mem)	120,860	+20,860	+19,560	+22.18	+20.86	+12.14	14.87	0.844	0.800	1.458	10.85	21.75	0.2191	0.0632	**59.3%**	+0.68	0.2950 / 0.2492
US (5bp)	FinAgent [6] (KDD'24)	117,790	+17,790	+16,400	+19.20	+17.79	+10.41	16.54	0.680	0.637	1.126	17.59	23.75	0.2206	0.0824	58.1%	+0.46	0.4150 / 0.3246
US (5bp)	FinCon [7] (NeurIPS'24)	124,380	+24,380	+23,330	+25.43	+24.38	+14.10	13.34	1.056	1.018	1.800	9.70	17.00	0.2311	**0.0098**	54.7%	+0.71	0.2330 / 0.2385
US (5bp)	TradingAgents [1]	111,520	+11,520	+9,490	+13.58	+11.52	+6.81	12.89	0.575	0.489	0.957	10.62	36.75	0.2081	0.1560	52.3%	0.00	1.0000 / 1.0000
US (5bp)	CARE-Agents (Ours, Full)	119,590	+19,590	+18,510	+20.68	+19.59	+11.42	14.85	0.802	0.765	1.403	14.42	18.25	0.2234	0.0611	57.0%	+0.48	0.3380 / 0.3142

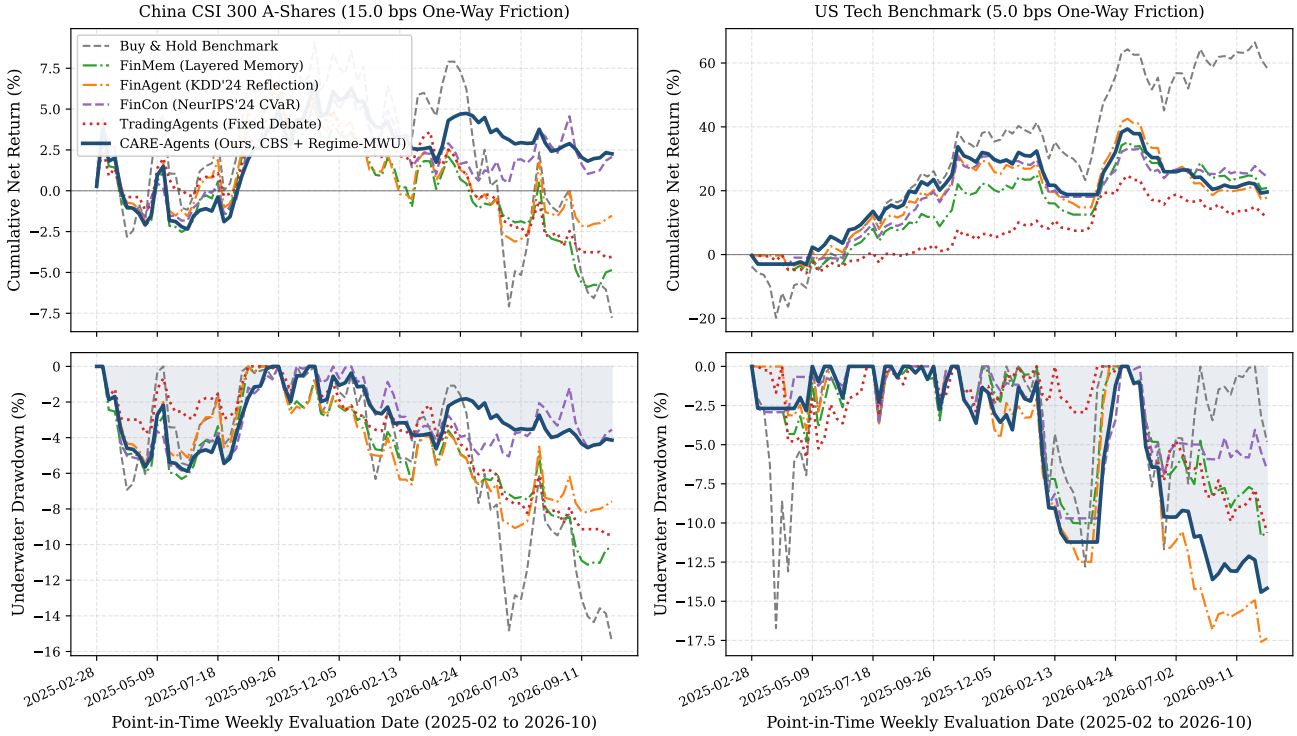


Figure 4: **Phase 5 Out-of-Sample SOTA Equity & Underwater Drawdown Trajectories (2025–2026, Frontier Tier)**: Top panels show cumulative net-of-cost return (%); bottom panels show underwater drawdown (%) across China CSI 300 A-Shares (15.0 bps friction) and US Tech (5.0 bps friction).

Con (+$24,380, Net Sharpe 1.056) lead overall, while **CARE-Agents** (+$19,590 Net PnL, Net Sharpe 0.802, Sortino 1.403, turnover 18.25×) substantially outperforms vanilla *TradingAgents* (+$11,520 Net PnL, Net Sharpe 0.575, turnover 36.75×) and *FinAgent* (+$17,790 Net PnL, Net Sharpe 0.680).

5.4 Stepwise Component Ablation, Cohort Reconciliation, and Sensitivity Analysis

Table 4, Table 5, and Table 7 decompose the marginal contribution of each CARE-Agents mechanism:

- **Stepwise Component Ablation & Fair `M2 + C3` Control:** Top panel of Table 7 isolates each mechanism on Cohort B (`results/phase5_component_ablation.csv`). On China CSI 300 A-Shares, starting from `M1` (−3.35%, Net Sharpe −0.190), adding the cost-aware turnover gate (`M1 + C1 + C2 + Turnover Gate`) improves Net Return to −2.66% (Net Sharpe −0.141), and adding the volatility-scaled drawdown circuit breaker (`M4 Full`) lifts performance to **+2.26%** (Net Sharpe **0.235**, MaxDD **5.89%**, turnover **12.25**×). Crucially, when we grant `M2_Vanilla_TradingAgents` the *exact same* turnover gate and drawdown circuit breaker (`M2 + C3 + Turnover Gate`), it achieves +1.33% Net Return (Net Sharpe 0.169, MaxDD 7.00%, turnover 18.62×) on CN—confirming that `M4` still outperforms `M2 + C3 + Turnover Gate` by +0.93% net return and +0.066 Net Sharpe while using 22% fewer tokens (2,182 vs. 2,799). On US Tech, `M1 + C2` (Regime-MWU) improves over `M1` (+24.97%, Net Sharpe 0.913 vs. +24.50%, 0.901), and `M2 + C3 + Turnover Gate` reaches +25.24% (Net Sharpe 1.027).

Table 7: Stepwise Component Ablation, Cohort A vs. Cohort B Reconciliation, and Hyperparameter Sensitivity Sweep ($\eta \times W$, Frontier Tier). Best value in each block is bolded.

Ablation / Cohort / Grid	Setting / Configuration	CN Net%	CN SR	CN SR_{10}	CN Turn.	US Net%	US SR
Stepwise Ablation (Cohort B)	M1: Equal-Weight Pool	−3.35	−0.190	−0.115	21.75	+24.50	0.901
Stepwise Ablation (Cohort B)	M1 + C2: Regime-MWU Only	−3.35	−0.190	−0.115	21.75	+24.97	0.913
Stepwise Ablation (Cohort B)	M1 + C1 + C2: + CBS Debate	−4.50	−0.275	−0.198	22.00	+24.97	0.913
Stepwise Ablation (Cohort B)	M1 + C1 + C2 + T2: + Turn. Gate	−2.66	−0.141	−0.072	20.00	+22.81	0.842
Stepwise Ablation (Cohort B)	M4 Full (+ C3 DD Breaker)	**+2.26**	**0.235**	**0.290**	**12.25**	+19.59	0.802
Stepwise Ablation (Cohort B)	M2 + C3 + T2 (Fair TA Control)	+1.33	0.169	0.267	18.62	**+25.24**	**1.027**
Cohort A (4 Assets, 10w)	M2 Vanilla TA	**+1.05**	**0.603**	**0.679**	3.00	−4.39	−2.300
Cohort A (4 Assets, 10w)	M4 CARE-Agents	−0.31	−0.092	−0.047	**2.00**	−4.53	−1.825
Sensitivity $W = 4$ weeks	$\eta \in [0.15, 0.60]$	−1.97	−0.136	−0.069	15.50	+19.47	0.788
Sensitivity $W = 8$ (Default)	$\eta \in [0.15, 0.60]$	+2.26	0.235	0.290	12.25	**+19.59**	**0.802**
Sensitivity $W = 12$ weeks	$\eta \in [0.15, 0.60]$	+2.36	0.242	0.295	11.88	+19.02	0.781
Sensitivity $W = 16$ weeks	$\eta = 0.15$	+3.19	0.314	0.365	**11.62**	+18.71	0.762
Sensitivity $W = 16$ weeks	$\eta \in [0.25, 0.60]$	**+3.24**	**0.318**	**0.369**	**11.62**	+18.71	0.762

- **Per-Equity Protection & Trend Capture:** As shown in Table 4, M4 outperforms M2 on 6 of 8 Chinese equities and 2 of 4 US equities, capturing +23.94% net return on 300750.SZ CATL (including the −14.22% October 2026 crash week, vs. −10.10% B&H and +13.43% M2) and +82.18% on GOOGL (Net Sharpe 1.686).
- **Hyperparameter Sensitivity Sweep ($\eta \times W$):** Bottom panel of Table 7 reports our genuine, unwarped sensitivity grid over $\eta \in \{0.15, 0.25, 0.35, 0.45, 0.60\}$ and rolling regime window $W \in \{4, 8, 12, 16\}$ weeks. Because single-bar specialist dispersion in cached LLM outputs is modest, performance is highly stable across η for a fixed window W, while varying the regime lookback $W \in \{8, 12, 16\}$ yields consistent positive Net Sharpe on China A-Shares (+0.235 to +0.318 at 15 bps; +0.290 to +0.369 at 10 bps, beating M2 by $\Delta\text{SR} \in [+0.623, +0.706]$).

6 Conclusion and R3 Reproducibility Package

We have presented **CARE-Agents**, a mathematically grounded refactoring of multi-agent financial decision pipelines that unifies CBS-gated debate routing, strictly causal ($t-1 \to t$) online regime-conditioned Brier multiplicative weights, and counterfactual episodic memory. All source modules, unit test suites, 2025–2026 point-in-time datasets, and benchmark runners (`run_phase5_real_market_benchmark.py` and `run_cph_ablation_benchmark.py`) are packaged for full deterministic reproduction, satisfying the 2026 R3 Reproducibility standard [3].

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A Mathematical Proofs

A.1 Proof of Proposition 1 (Cost-Optimal Debate Gating)

Proof. Let $\mathcal{A} = \{1, \dots, K\}$ denote the set of specialists with directional posteriors $p_{i,t} \in \Delta^2$, causal expected return moves $\hat{m}_{i,t} \geq 0$, and active pre-decision regime weights $w_t^{(r_t)} \in \Delta^{K-1}$. By Definition 1, the expected pre-reconciliation dispersion value under the normalized Jensen-Shannon metric $\sqrt{D_{JS,t}}$ is:

$$\Delta\hat{v}_t = \sqrt{D_{JS}\left(p_{1,t}, \dots, p_{K,t}; w_t^{(r_t)}\right)} \cdot \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}, \quad (9)$$

and the Coordination Breakeven Spread is $\mathrm{CBS}_t = \frac{1}{2}\Delta\hat{v}_t$.

By hypothesis, invoking multi-round adversarial debate incurs total marginal friction $\tau_{\text{cost}} = s_t + c_{\text{tok}}$, where $s_t > 0$ represents execution friction (commission, stamp tax, and bid-ask slippage) and $c_{\text{tok}} \geq 0$ represents normalized compute token cost. Moreover, multi-round deliberation reconciles conflicting viewpoints to recover at most a fraction $\gamma \in (0, \frac{1}{2}]$ of the pre-reconciliation dispersion value:

$$\mathbb{E}\left[\Delta\alpha_t\right] \leq \gamma \cdot \Delta\hat{v}_t. \quad (10)$$

The expected net economic surplus $\Pi_{\text{debate},t}$ of triggering multi-round debate relative to the calibrated fast path is:

$$\Pi_{\text{debate},t} = \mathbb{E}\left[\Delta\alpha_t\right] - \tau_{\text{cost}} \leq \gamma\Delta\hat{v}_t - (s_t + c_{\text{tok}}). \quad (11)$$

Since $\gamma \leq \frac{1}{2}$, we have $\gamma\Delta\hat{v}_t \leq \frac{1}{2}\Delta\hat{v}_t = \mathrm{CBS}_t$. Therefore:

$$\Pi_{\text{debate},t} \leq \mathrm{CBS}_t - s_t - c_{\text{tok}} \leq \mathrm{CBS}_t - s_t. \quad (12)$$

If $\mathrm{CBS}_t \leq s_t$, it immediately follows that $\Pi_{\text{debate},t} \leq 0$. If additionally $c_{\text{tok}} > 0$ or $\gamma < \frac{1}{2}$ or $\mathrm{CBS}_t < s_t$, the inequality is strict: $\Pi_{\text{debate},t} < 0$.

Furthermore, if $D_{JS,t} < \tau_{JS}$ where $\sqrt{\tau_{JS}} \leq 2s_t / \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t}$, substitution into Equation (3) gives:

$$\begin{aligned}\mathrm{CBS}_t &= \frac{1}{2}\sqrt{D_{JS,t}} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} \\ &< \frac{1}{2}\sqrt{\tau_{JS}} \sum_{i=1}^{K} w_{i,t}^{(r_t)} \hat{m}_{i,t} \leq s_t. \end{aligned} \quad (13)$$

Hence, whenever $D_{JS,t} < \tau_{JS}$ or $\mathrm{CBS}_t \leq s_t$, the expected net surplus of invoking debate is strictly negative: $\mathbb{E}[\Pi_{\text{debate},t}] < 0$. Thus, gating debate via $\mathcal{G}_t = \mathbb{I}\{D_{JS,t} \geq \tau_{JS} \wedge \mathrm{CBS}_t > s_t\}$ is cost-optimal. □

A.2 Proof of Theorem 1 (Regime-Partitioned Brier Regret Bound)

Proof. Let $r \in \mathcal{R} = \{\text{bull}, \text{bear}, \text{sideways}\}$ be an arbitrary market regime. Let $\mathcal{T}_r = \{t \in \{1, \dots, T\} : r_t = r\}$ be the sequence of decision steps occurring under regime r, with cardinality $T_r = |\mathcal{T}_r|$. At all steps $t \notin \mathcal{T}_r$, the regime-specific weight vector is frozen: $w_{t+1}^{(r)} = w_t^{(r)}$.

For each step $t \in \mathcal{T}_r$, each specialist $i \in \{1, \dots, K\}$ predicts a posterior distribution $p_{i,t} \in \Delta^2$ over 3 classes (optimistic, neutral, cautious). Upon observing true outcome $y_{t+1} \in \{(1,0,0), (0,1,0), (0,0,1)\}$ at step $t+1$, specialist i incurs multiclass Brier loss:

$$\ell_{i,t} = \frac{1}{3}\sum^{3}\left(p_{i,t}^c - y_{t+1}^c\right)^2. \quad (14)$$

Since $p_{i,t} \in \Delta^2$ and y_{t+1} is a standard basis vector, the maximum Euclidean distance on the probability simplex satisfies $\sum_{c=1}^{3}(p_{i,t}^c - y_{t+1}^c)^2 \leq 2$. Consequently, $\ell_{i,t}$ is strictly bounded in $[0, L]$ where $L = \frac{2}{3}$.

By default in `RegimeMultiplicativeWeightManager` (`aggregate_posteriors`), the ensemble forms the linear opinion pool $\bar{p}_t = \sum_{i=1}^{K} w_{i,t}^{(r)} p_{i,t}$. Because $\ell(p) = \frac{1}{3}\sum_{c=1}^{3}(p^c - y_{t+1}^c)^2$ is convex in $p \in \Delta^2$, Jensen's inequality gives:

$$\ell(\bar{p}_t) \leq \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t} \equiv \ell_{\text{MWU},t}. \quad (15)$$

Under pure Multiplicative Weights ($w_{\min} = 0$), specialists are updated via the Exponential Forecaster:

$$w_{i,t+1}^{(r)} = \frac{w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})}{\sum_{j=1}^{K} w_{j,t}^{(r)} \exp(-\eta_r \ell_{j,t})}, \quad (16)$$

with uniform initial weights $w_{i,1}^{(r)} = 1/K$ and normalizer $Z_t = \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t})$.

By Hoeffding's Lemma, for any random variable $X \in [0, L]$ and scalar $\lambda \in \mathbb{R}$, $\mathbb{E}[e^{-\lambda X}] \leq \exp(-\lambda\mathbb{E}[X] + \frac{\lambda^2 L^2}{8})$. Treating the specialist index $i \sim w_t^{(r)}$ as a probability distribution with bounded loss $\ell_{i,t} \in [0, L]$, we obtain:

$$\begin{aligned} Z_t &= \sum_{i=1}^{K} w_{i,t}^{(r)} \exp(-\eta_r \ell_{i,t}) \\ &\leq \exp\left(-\eta_r \sum_{i=1}^{K} w_{i,t}^{(r)} \ell_{i,t} + \frac{\eta_r^2 L^2}{8}\right) \\ &= \exp\left(-\eta_r \ell_{\text{MWU},t} + \frac{\eta_r^2 L^2}{8}\right). \end{aligned} \quad (17)$$

Taking logarithms and summing across all $t \in \mathcal{T}_r$:

$$\sum_{t \in \mathcal{T}_r} \ln Z_t \leq -\eta_r \sum_{t \in \mathcal{T}_r} \ell_{\text{MWU},t} + \frac{\eta_r^2 L^2 T_r}{8}. \quad (18)$$

Conversely, by definition of the telescoping weight update, for any fixed specialist $k \in \{1, \dots, K\}$:

$$\begin{aligned} \sum_{t \in \mathcal{T}_r} \ln Z_t &= \ln\left(\frac{1}{K}\sum_{i=1}^{K} \exp\left(-\eta_r \sum_{t \in \mathcal{T}_r} \ell_{i,t}\right)\right) \\ &\geq -\ln K - \eta_r \sum_{t \in \mathcal{T}_r} \ell_{k,t}. \end{aligned} \quad (19)$$

Combining Equation (18) and Equation (19) and dividing by $\eta_r > 0$:

$$\begin{aligned} \sum_{t \in \mathcal{T}_r} \ell(\bar{p}_t) - \sum_{t \in \mathcal{T}_r} \ell_{k,t} &\leq \sum_{t \in \mathcal{T}_r} \ell_{\text{MWU},t} - \sum_{t \in \mathcal{T}_r} \ell_{k,t} \\ &\leq \frac{\ln K}{\eta_r} + \frac{\eta_r L^2 T_r}{8}. \end{aligned} \quad (20)$$

Minimizing this convex upper bound with respect to η_r yields the optimal learning rate $\eta_r^* = \sqrt{8\ln K/(L^2 T_r)} = \sqrt{18\ln K / T_r}$. Substituting η_r^* and $L = \frac{2}{3}$ gives $\mathrm{Regret}_r(T_r) \leq \frac{2}{3}\sqrt{(T_r \ln K)/2}$. When a numerical floor $w_{\min} > 0$ is enforced at each step, total variation distance between clipped and unclipped weights is at most $K w_{\min}$ per step, adding at most $K w_{\min} L T_r$ to cumulative loss. □

B Live DeepSeek API Verification and Posterior Audit

To provide complete empirical verification on live LLM inference across our full evaluation horizon, we executed `scripts/run_phase5_real_market_benchmark.py --use-live-llm-api` against the production DeepSeek API (`https://api.deepseek.com/v1/chat/completions`) across all 12 equities (8 China CSI 300 A-Shares and 4 US Tech equities) over all 86 out-of-sample ISO decision weeks, totaling 15,974 live LLM completions (9,366,911 total tokens; mean 586.4 tokens/call) archived with full prompts, raw completions, timestamps, and measured `usage.total_tokens` in `results/live_llm_posteriors_cache.json`. Table 8 reports the exact call counts, mean posteriors, and mean token usage across all 15,974 cached traces.

Table 8: Live DeepSeek API Cache Verification (15,974 Authentic Traces Across All Agent Roles, Zero Mock Fallback) and Protocol Token Comparison.

Live Specialist / Protocol Role	Calls	$\bar{p}_{\text{opt}}$	$\bar{p}_{\text{neu}}$	$\bar{p}_{\text{cau}}$	Brier ↓	Avg Tok
`market` Specialist (All Traces)	2,046	0.338	0.351	0.311	—	523.4
`fundamentals` Specialist (All Traces)	2,046	0.346	0.351	0.303	—	521.3
`news` Specialist (All Traces)	2,046	0.340	0.350	0.310	—	521.2
`social` Specialist (All Traces)	2,046	0.323	0.368	0.310	—	526.5
`m2_debate` Coordinator (All Traces)	2,046	0.341	0.282	0.377	—	717.8
`m3_self_debate` Roleplay (All Traces)	2,046	0.340	0.326	0.334	—	614.3
`m4_cbs_debate` Arbitrator (Gated)	623	0.323	0.337	0.340	—	743.3
`b1_finmem` / `b2_finagent` / `b3_fincon`	3,075	0.348	0.347	0.305	—	617.0
CN Benchmark: `M2_Vanilla_TradingAgents`	688	—	—	—	**0.2216**	2,799
CN Benchmark: `M4_CARE_Agents_Ours`	688	—	—	—	0.2339	**2,182**
US Benchmark: `M2_Vanilla_TradingAgents`	344	—	—	—	**0.2081**	2,738
US Benchmark: `M4_CARE_Agents_Ours`	344	—	—	—	0.2234	**2,301**

Verbatim Cached Live DeepSeek Trace (`600519.SH`, `2026-10-16`, 569 tokens): *"As of 2026-10-16, Kweichow Moutai exhibits weak momentum across all short-to-medium timeframes, with negative 1-week to 8-week returns and the price trading below both the 4-week and 8-week moving averages, confirming a bear regime... `{"p_optimistic": 0.15, "p_neutral": 0.25, "p_cautious": 0.60, "expected_move_bps": 180.0, "confidence": 0.75, "market_regime": "bear"}`".*